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Land Cover Classification in the Kaveri Region Using Landsat-8 and Google Earth Engine

1. Introduction

The increasing demand for precise and timely information on Land Cover (LC) change is paramount for effective resource management. This study addresses the need for automated LC mapping in the Kaveri River Region, India, an area of significant ecological and agricultural importance, by leveraging satellite remote sensing and advanced machine learning techniques.

This report details a complete methodology implemented within the Google Earth Engine (GEE) platform, from preprocessing Landsat-8 imagery and preparing a robust training dataset to training and evaluating a Random Forest (RF) classifier. The successful application of this workflow achieved high classification accuracy, offering crucial insights for water resource management, urban planning, and environmental monitoring in the basin.

2. Study Area

The study focuses on the **Kaveri River Region** in Southern India. The geographical extent of the basin was defined by importing a pre-existing geometry asset into GEE and simplifying it to optimize computational efficiency.

GEE Code Snippet (Study Area Definition):

```
var studyarea = ee.FeatureCollection('projects/ganga-region-surface-water-1/assets/Kaveri_River').geometry().simplify(1500);
```

3. Data Source and Preprocessing

3.1. Data Collection: Landsat-8

The project utilized the USGS Landsat-8, Collection 2, Tier 1, Surface Reflectance (SR) and Surface Temperature (ST) products. The dataset was filtered to the study area and a specific time period: January 1, 2020, to January 1, 2024, ensuring only images with a cloud cover of less than 10% were included.

3.2. Preprocessing & Scaling

All images in the collection were subjected to scaling functions to convert the raw Digital Numbers (DN) into standardized physical units (reflectance and Kelvin).

- **Optical Bands (SR_B*)**: Scaled by 0.0000275 and offset by -0.2.
- **Thermal Bands (ST_B*)**: Scaled by 0.00341802 and offset by 149.0.

GEE Code Snippet (Scaling Function):

```
var optical = image.select('SR_B.*').multiply(0.0000275).add(-0.2).float();

var thermal = image.select('ST_B.*').multiply(0.00341802).add(149.0).float();
```

3.3. Median Composite

To effectively mitigate noise from atmospheric variability and sporadic cloud cover across the time series, a median composite was generated from the entire filtered and rescaled image collection. This resultant single-image mosaic formed the basis for all subsequent classification steps.

4. Methodology: Land Cover Classification

4.1. Predictor Variable Selection

Seven spectral bands were selected from the processed median composite to serve as the input properties (predictors) for the machine learning model. These bands cover the visible, near-infrared, shortwave-infrared, and thermal spectrum, crucial for discriminating between various LC types.

Band Name	Description
SR_B2	Blue
SR_B3	Green
SR_B4	Red
SR_B5	Near-Infrared (NIR)
SR_B6	Shortwave Infrared 1 (SWIR-1)
SR_B7	Shortwave Infrared 2 (SWIR-2)

Band Name	Description
ST_B10	Thermal Infrared (TIR)

4.2. Training Data Preparation

The final workflow utilized a pre-collected and exported training dataset (likely sampled via GEE's sampling function and stored as a CSV asset). This approach ensures a consistent and controlled dataset for training.

1. **Data Loading:** The dataset, containing 597 total training points (features with all selected band values and a mandatory '**Class**' property), was loaded directly from a GEE Asset.
2. **Train/Test Split:** The loaded data was randomly partitioned to create the sets required for training and validation:
 - o **Training Set (80%):** 467 features.
 - o **Testing Set (20%):** 130 features.

4.3. Random Forest Classifier Implementation

The Random Forest (RF) algorithm, known for its robustness against overfitting and ability to handle high-dimensional data, was selected for the classification. The model was configured with 10 trees (`ee.Classifier.smileRandomForest(10)`).

The classifier was trained using the 467 features in the `trainSet`, predicting the '**Class**' property based on the seven defined bands.

Land Cover Classes and Color Palette:

- **0 - Waterbody** (Dark Blue: #0c2c84)
- **1 - Urban Area** (Red: #e31a1c)
- **2 - Vegetation** (Dark Green: #005a32)
- **3 - Barren Land** (Light Yellow: #fed976)

5. Results and Analysis

5.1. Classification and Map Output

The trained RF classifier was applied to the entire median composite image, producing a wall-to-wall Land Cover map of the Kaveri River Region. The final classified image was exported in GeoTIFF format for further geospatial analysis and also generated as a high-resolution PNG for visual assessment.

5.2. Accuracy Assessment

The performance of the RF model was rigorously assessed using the **Testing Set (130 points)** to generate an **Error Matrix (Confusion Matrix)**.

Overall Accuracy

The model demonstrated a high level of performance:

Overall Classification Accuracy: 81.53%

The matrix is read (Actual Class on Rows, Predicted Class on Columns). The order of classes is 0 (Waterbody), 1 (Urban), 2 (Vegetation), 3 (Barren Land).

Producer's and Consumer's Accuracies

The class-specific accuracies provide deeper insights into the model's reliability:

Class	Producer's Accuracy (Completeness)	Consumer's Accuracy (Purity)
Waterbody (0)	25 / 31 approx 80.65%	25 / 31 approx 80.65%
Urban Area (1)	27 / 28 approx 96.43%	27 / 33 approx 81.82%
Vegetation (2)	34 / 41 approx 82.93%	34 / 39 approx 87.18%
Barren Land (3)	20 / 30 approx 66.67%	20 / 25 approx 80.00%

The results show **Urban Area** was classified with the highest Producer's Accuracy (high completeness), while **Vegetation** had the highest Consumer's Accuracy (high purity). The **Barren Land** class showed the lowest Producer's Accuracy, indicating it was the class most often confused with others (likely Waterbody and Vegetation in the training data).

5.3. Variable Importance Analysis

The Random Forest model provides a key advantage: The Variable Importance metric, which ranks the predictive power of the input bands. This analysis confirms which spectral features are most critical for distinguishing the land cover classes.

Key Finding: The **Near-Infrared (SR_B5)** band is consistently the most important variable in Land Cover classification due to its strong sensitivity to vegetation health and biomass.

The complete variable importance can be visualized in the chart generated by the GEE script.

6. Significance and Conclusion

Significance

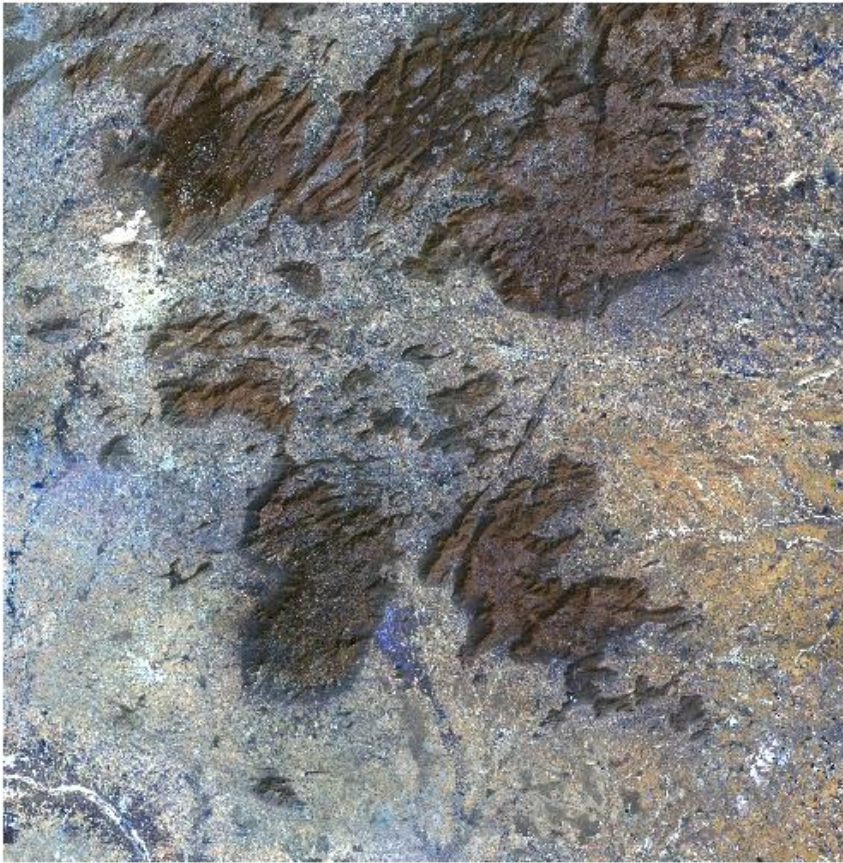
- **Scalability:** The use of GEE provides a low-cost, scalable framework that bypasses local computational limitations, making the workflow highly transferable to other large river basins or data-scarce regions.
- **Integration of ML:** The project successfully demonstrates the integration of satellite remote sensing with the Random Forest machine learning framework for robust environmental analysis.
- **Actionable Outputs:** The highly accurate Land Cover map is immediately applicable to urban planning, agriculture monitoring, and crucial water resource management within the Kaveri basin.

Conclusion

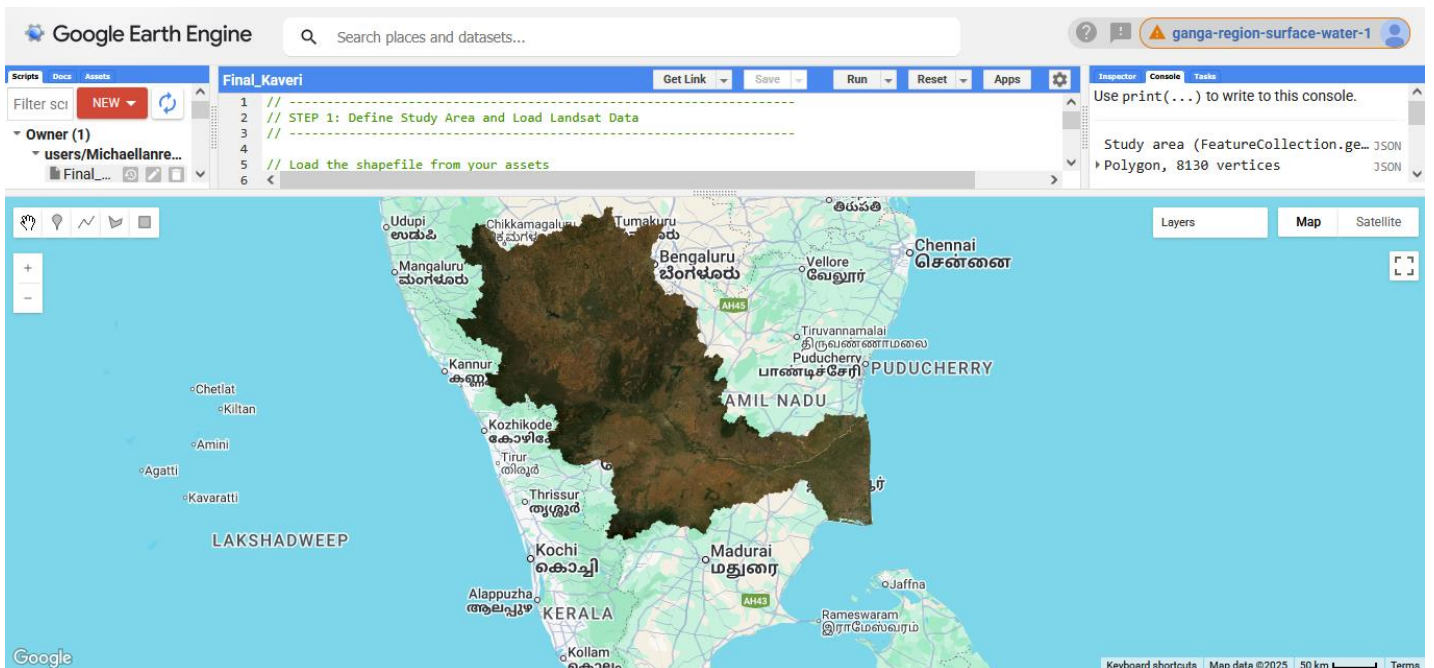
This study successfully implemented a comprehensive and reproducible workflow in Google Earth Engine, leveraging Landsat-8 data and the Random Forest classifier to produce an accurate Land Cover map of the Kaveri River Region with an **Overall Accuracy of 81.53%**. This work highlights the potential of cloud-based platforms and machine learning to deliver rapid, accurate, and scalable environmental monitoring solutions where conventional field surveys are often impractical.

RESULT

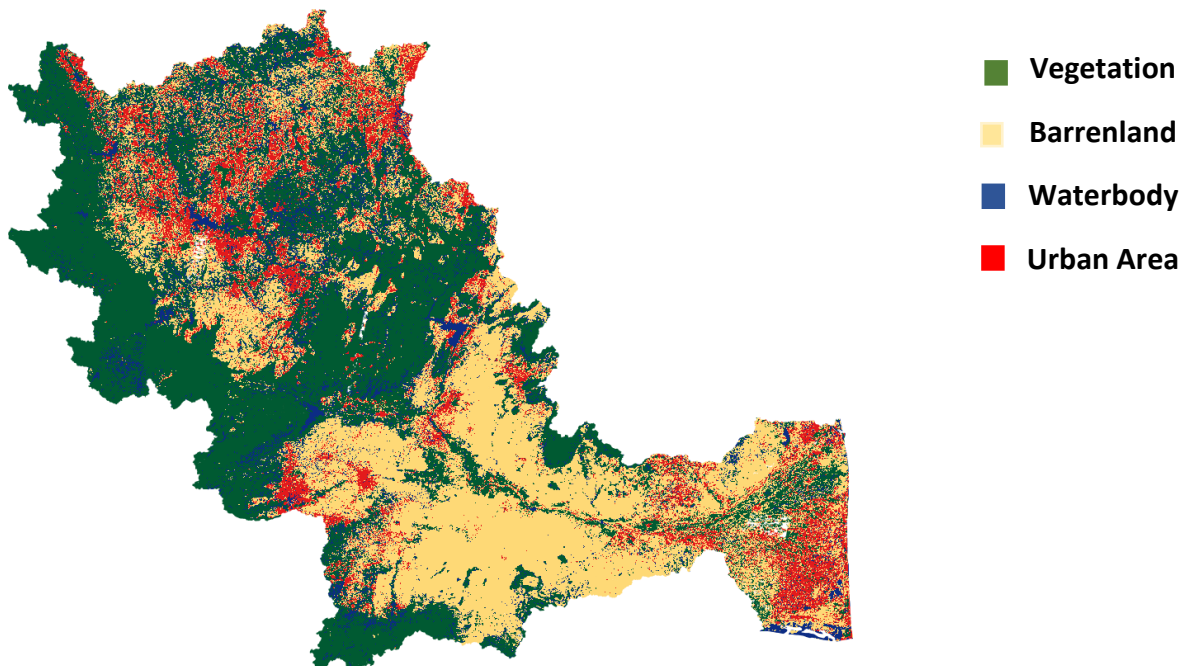
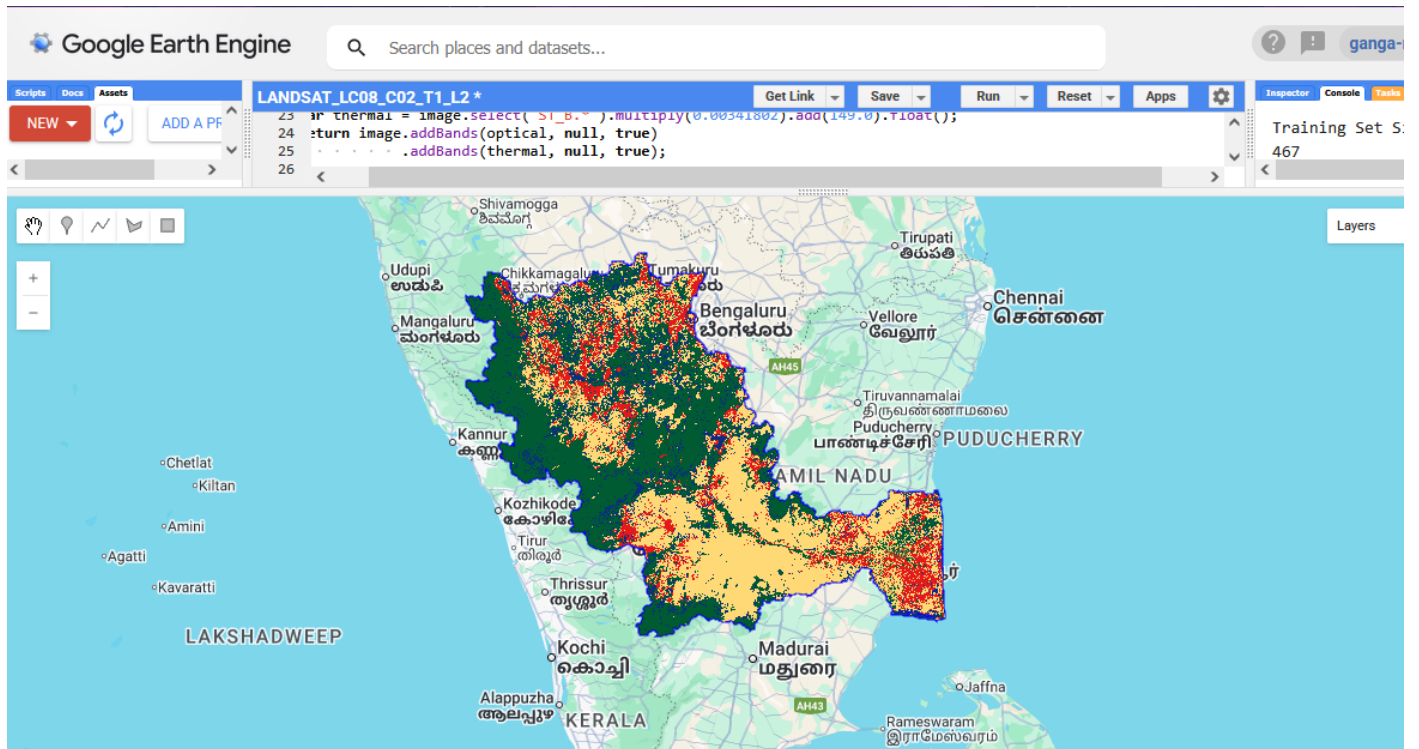
A SECTION OF THE RAW SURFACE REFLECTANCE IMAGE OF THE KAVERI REGION

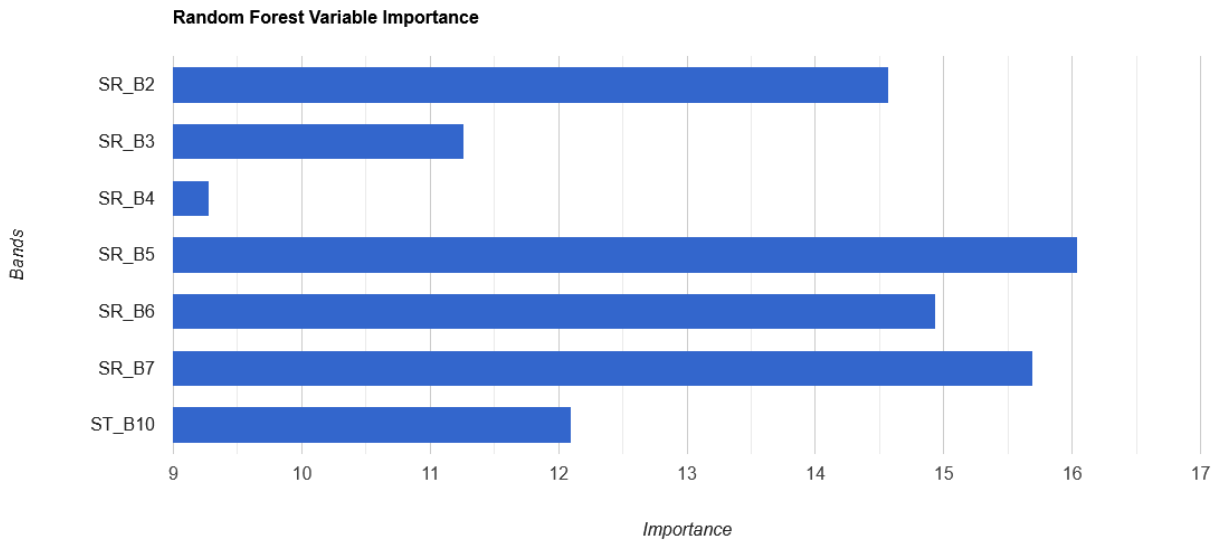


Display of a section of the Kaveri River Basin



Landcover Classification of the Kaveri River Basin





Google Earth Engine

Search places and datasets...

ganga-region-surface-water-1

Scripts Docs Assets

NEW ADD A PROJECT

CLOUD ASSETS

- ganga-region-surface-w...
- CNN_Training_Data
- Kaveri_River
- Lake_Victoria_Shapefile
- Vectoriz
- Vectorized
- nigeria_The_Federal_R...

LEGACY ASSETS

- users/Michaelanrey
- This folder is empty.

LAN... Get Link Save Run Reset Apps

```
1
2 STEP 1: Define Study Area (No Change)
3
4 table = ee.FeatureCollection('projects/ganga-region-sur
5 studyarea = table.geometry(); // Full geometr
6 studyFeatureGeom = table.first().geometry(); // First f
7
8 Enter map manually
9 .setCenter(77.5, 12.0, 7);
10 .addLayer(studyarea, {color: 'blue'}, 'Study Area');
11
12
13 STEP 2: Load Landsat-8 Collection (No Change)
14
15 landsat8 = ee.ImageCollection("LANDSAT/LC08/C02/T1_L2")
16 .filterBounds(studyarea)
17 .filterDate('2020-01-01', '2024-01-01')
18 .filter(ee.Filter.lt('CLOUD_COVER', 10));
19
20 Apply scaling factors to Landsat-8
21 function applyScale(image) {
22   var optical = image.select('SR_B.*').multiply(0.0000275);
23   var thermal = image.select('ST_B.*').multiply(0.00341802);
24   return image.addBands(optical, null, true)
25     .addBands(thermal, null, true);
26
27
28 rescaled = landsat8.map(applyScale);
29 image = rescaled.median().clip(studyarea);
30
31 .addLayer(image, {bands:['SR_B4','SR_B3','SR_B2'], min:0
32
33 Define bands for classification
34 bands = ['SR_B2','SR_B3','SR_B4','SR_B5','SR_B6','SR_B7
35
36
```

Inspector Console Tasks

Use print(...) to write to this console.

Total Training Points from CSV: 597 JSON

Training Set Size: 467 JSON

Testing Set Size: 130 JSON

Confusion Matrix: JSON

[[25,1,3,2],[0,27,1,5],[4,0,34,3],[2,0,3,20]] JSON

Overall Accuracy: 0.8153846153846154 JSON

Producers Accuracy: List (4 elements) JSON

Consumers Accuracy: List (1 element) JSON

Variable Importance: Feature JSON